

IMPERIAL

Company Technical Meeting

Group 19

G19

17/06/2026

Team Members

Agastya Mehta	Andrew Lee	Ethan Hewitt
Aerodynamics Code	Structures Code	Aerodynamics Code

Keith Chan	Liem Fairbairn	Sohoon Kim
CAD	Aerodynamics Code	CAD

Manufacturing Hours

Name	Laser-Cutting & 3D printing	Workshop	Total
Agastya	0	10	10
Andrew	4	10	14
Ethan	0	10	10
Keith	1	3	4
Liem	0	9	9
Sohoon	2	12	14

Turbine Summary

2 Blades

474mm design diameter (212mm per blade design)

S834 at the root & SG6040 near the tip

25g of PLA for root & tip, two 5mm width carbon fibre, foams, 2 sheets of 2mm plywood, epoxy

3D printer for root & tip, laser cut for the aerofoil sections, hot-wire cutter for foams, and then use wood glue and epoxy to stick all surfaces together

Blade Assembly

PART	MATERIAL	PART DRAWING NUMBER	QUANTITY
W1	Plywood (2 ply)	AERO40004-DMT-G19-003	14
W2	Plywood (2 ply)	AERO40004-DMT-G19-005	3
W3	Plywood (2 ply)	AERO40004-DMT-G19-007	2
F1	Styrofoam	AERO40004-DMT-G19-004	1
F2	Styrofoam	AERO40004-DMT-G19-006	1
F3	Styrofoam	AERO40004-DMT-G19-008	1
P1	PLA	AERO40004-DMT-G19-002	1
P2	PLA	AERO40004-DMT-G19-009	1
CF1	Carbon Fibre Strip	AERO40004-DMT-G19-010	1
CF2	Carbon Fibre Strip	AERO40004-DMT-G19-011	1

ALPHABETS DENOTE MATERIAL:

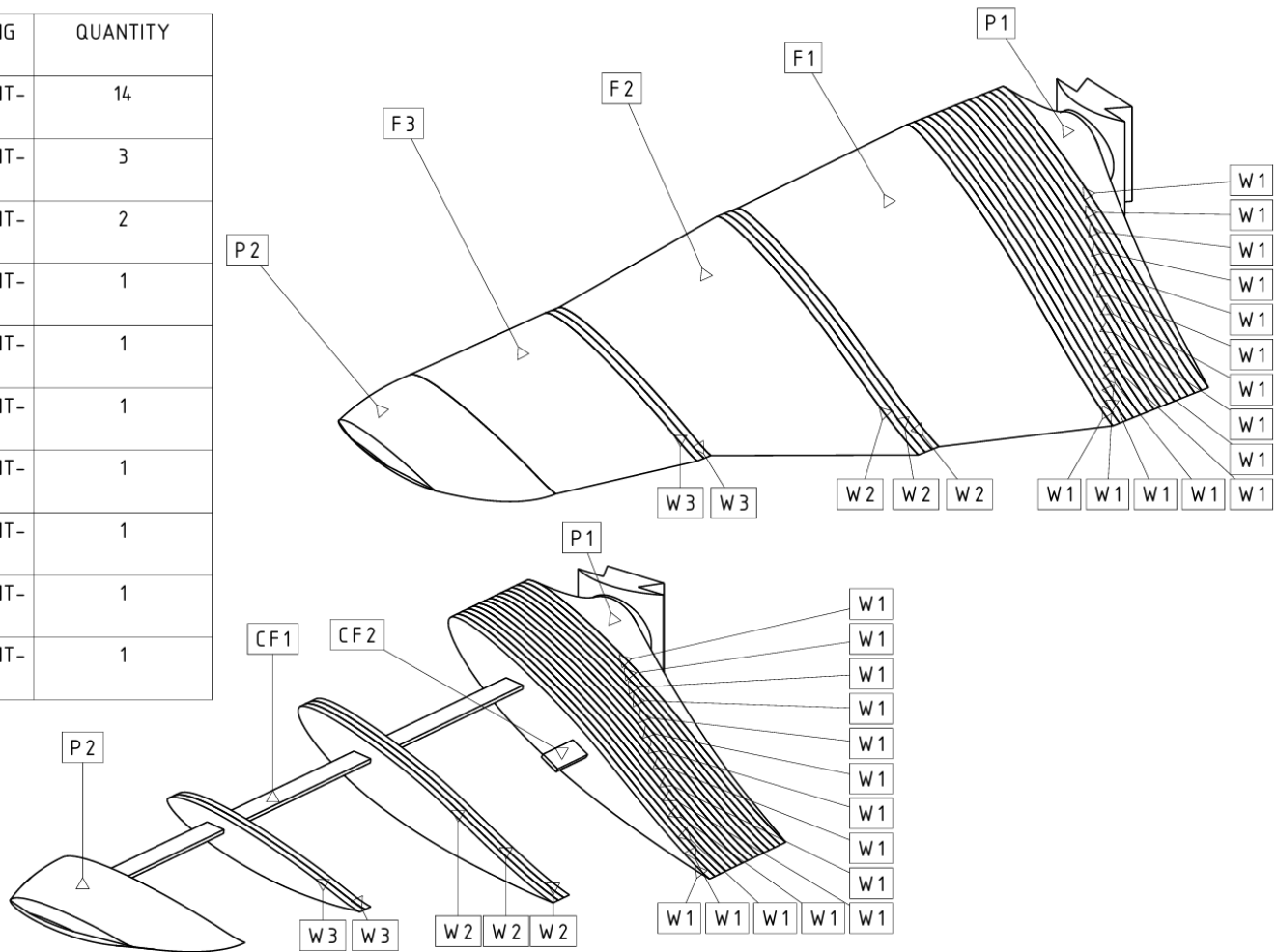
W: WOOD (PLYWOOD)

P: PLA

F: FOAM

CF: CARBON FIBRE

NUMBERS AFTER ALPHABET
DIFFERENTIATE PARTS OF SAME MATERIAL



TOLERANCE		THIRD ANGLE PROJECTION	MATERIAL:		TITLE:	Imperial College London
x=±0.5 x.X=±0.1 x.XX=±0.02	ANGULAR ±1°		DO NOT SCALE DRAWING		AE1 EP1 G19 DMT BLADE PROTOTYPE	
	NAME	DATE	ALL DIMENSIONS ARE IN MILLIMETERS		DRAWING NUMBER:	DEPARTMENT OF AERONAUTICS
DRAWN	G19	29/05/2026	A3 Scale 1:1		AERO40004-DMT-G19-001	SHEET 1 OF 2 REVISION A.1
CHECKED	G19	29/05/2026				
APPROVED	G19	29/05/2026				

Finished Blade

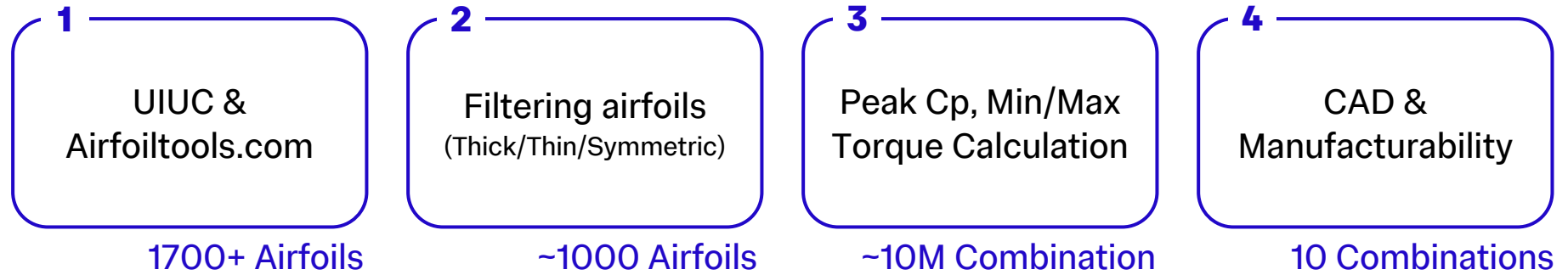
- Final Lengths: 209mm per blade.
- Masses: 46.33g & 45.82g



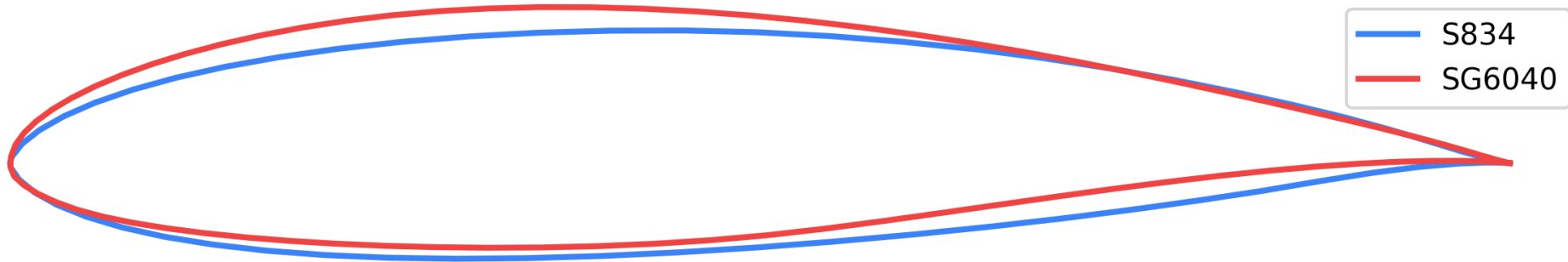
Presentation Overview

- Design Methodology including total cost
 - Airfoil Selection Process
 - Material Selection Process
- Aerodynamic Performance Prediction
- Structural Performance Prediction
- Prototype Manufacturing Methodology

Aerofoil Selection Process

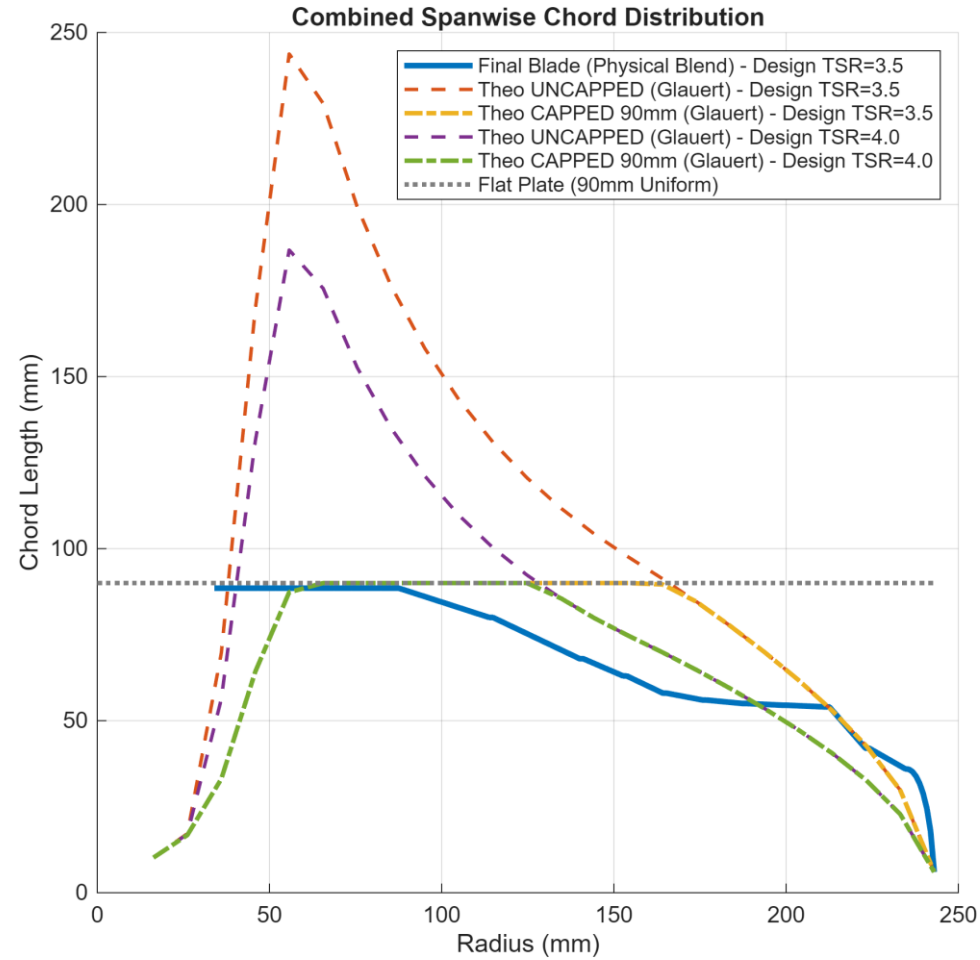


Final Selected Combination



Overall Design Methodology

Chord Distribution



Material Selection Process

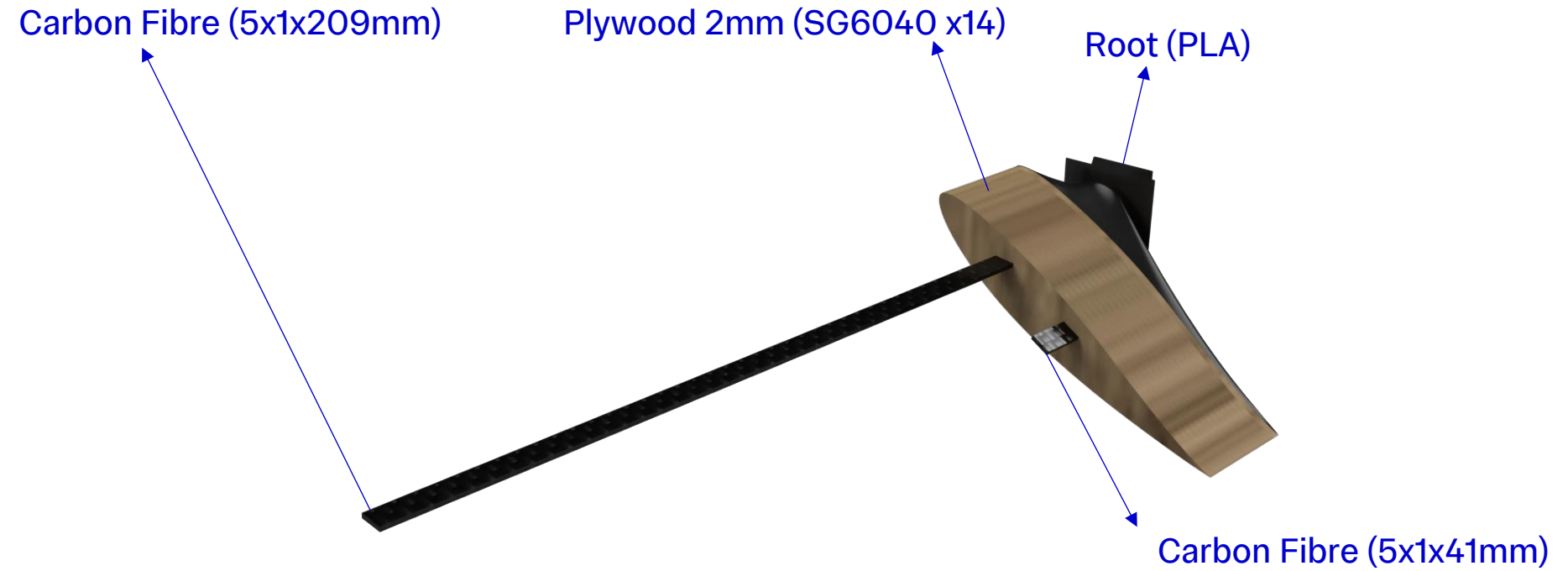
Comparison of each Material

Criteria	Styrofoam Block	Balsa Wood	Plywood Sheet	Acrylic sheet
Density	33 kg/m ³	140 kg/m ³	750 kg/m ³	1190 kg/m ³
Strength /10	3	6	8	8
Cost /10	10	8	5	5
Manufacturability /10	4	5	7	5
Total /30	17	19	20	18

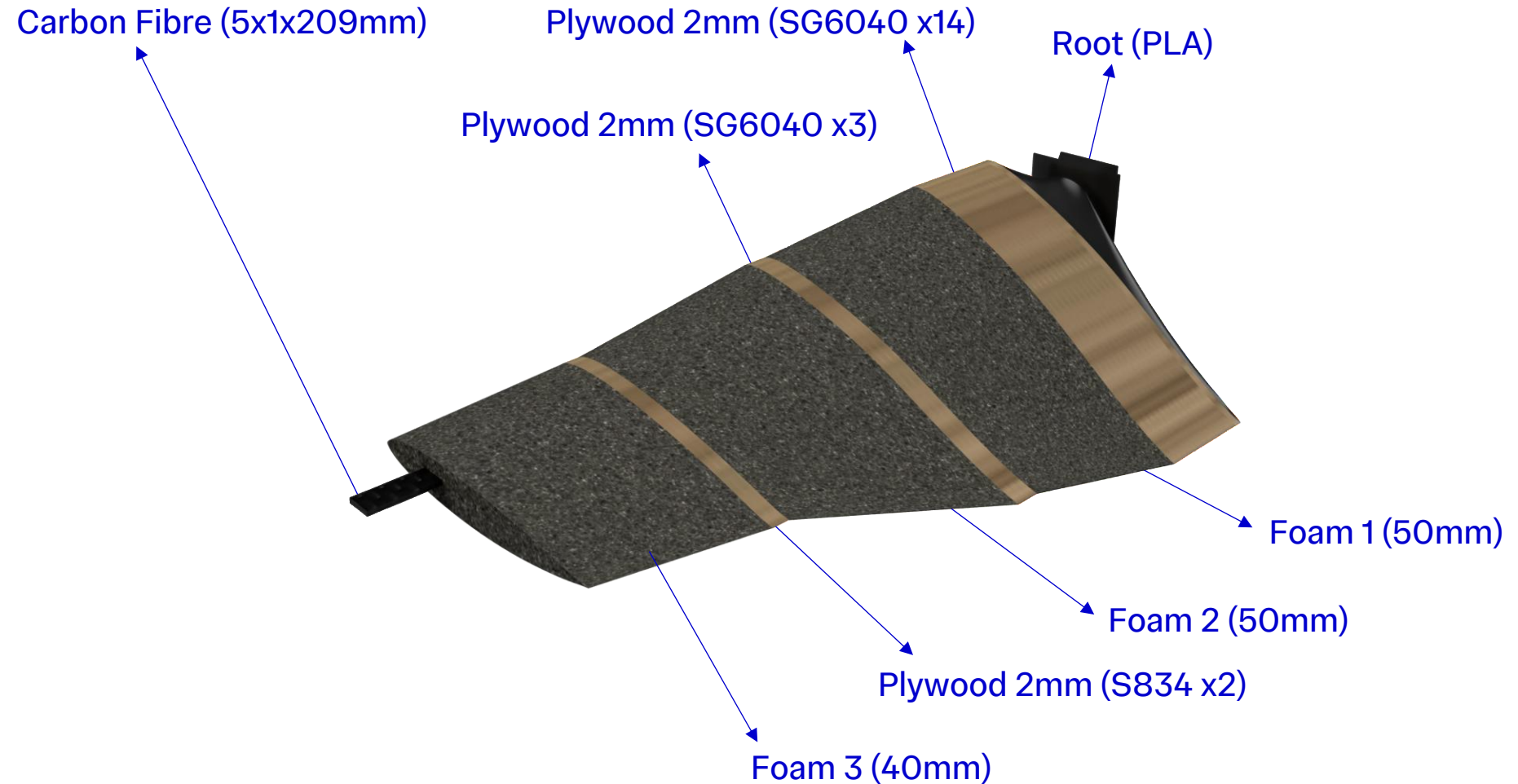
Possible Combinations

Criteria	Styrofoam + Plywood	Balsa Wood + Acrylic	Balsa Block + Plywood Sheet	Plywood Only (Reference)
Strength	6	?	7	8
Cost	8	7	7	5
Manufacturability	6	5	6	7
Total	20	?	20	20

Design Methodology



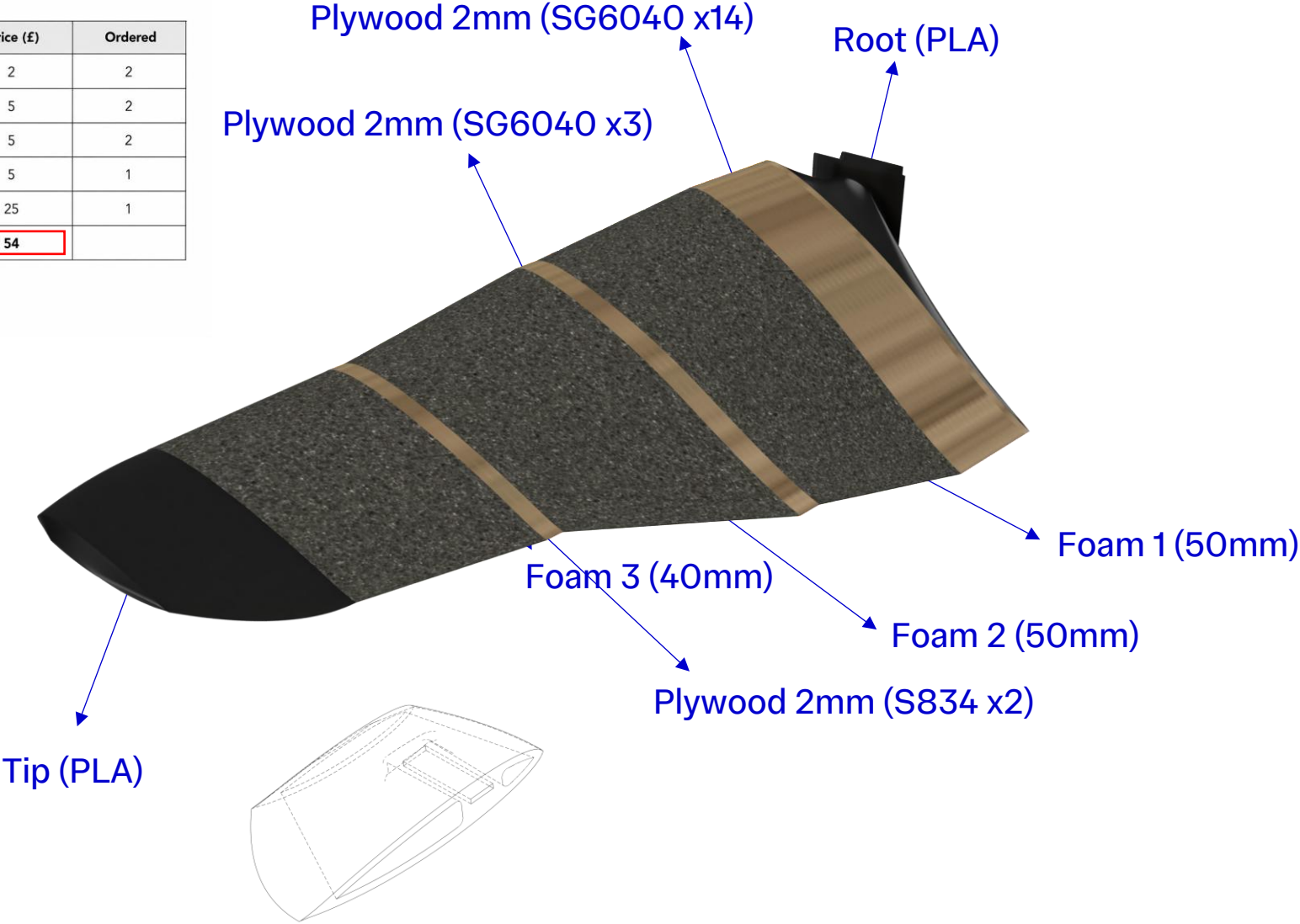
Design Methodology



Blade Assembly

With Cost

Item / Material	Price (£)	Ordered
Styrofoam Block	2	2
Plywood Sheet	5	2
Carbon Fibre Strip	5	2
Epoxy	5	1
PLA	25	1
Cost Spent	54	



Aerodynamic Performance Prediction

Equation Set and Extensions

- $\lambda_r = \lambda_{design} \cdot \left(\frac{r}{R}\right)$
- $c = \frac{16\pi R^2}{9B\lambda_{design}^2 r C_{l,des}}$
- $\beta = \arctan\left(\frac{2}{3\lambda_r}\right) - \alpha_{design}$
- $\phi = \arctan\left(\frac{1-a}{\lambda_r(1+a')}\right)$
- $a_{new} = \frac{\sigma C_n}{4F \sin^2 \phi + \sigma C_n}$
- $a'_{new} = \frac{\sigma C_t}{4F \sin \phi \cos \phi - \sigma C_t}$
- $F_{tip} = \frac{2}{\pi} \arccos\left(e^{-\frac{B(R-r)}{2r \sin \phi}}\right)$
- $a_{new} = \frac{18F - 20 - 3\sqrt{C_T(50 - 36F) + 12F(3F - 4)}}{36F - 50}$
- $C_{l,3D} = C_{l,2D} + 3\left(\frac{c}{r}\right)^2 \Delta C_l (1 - \mu)$
- $C_{D,90} = 1.11 + 0.018 \times AR$

Stall Delay Correction

$$C_{l,current} = C_{l,current} + \frac{3c^2}{r^2} \times \Delta C_l$$

Aerofoil Drag

Equations Accounting for drag:

$$\Delta D = 0.5\rho W^2 C_{Dc} \times Elmt Width$$

$$T_{axial} = B(\Delta L \times \cos(\varphi) - \Delta D \times \sin(\varphi))$$

$$F_{tangential} = B(\Delta L \times \sin(\varphi) - \Delta D \times \cos(\varphi))$$

Aerodynamic Performance Prediction

Aero Code Extensions

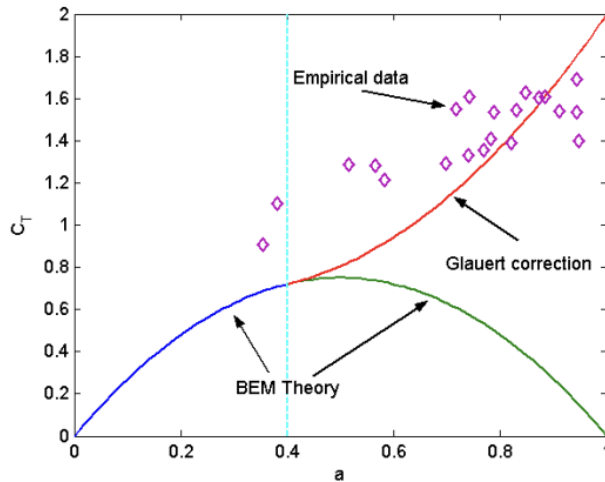
Glauert Optimum Rotor & Wake Rotation

$$\Delta p = 2\rho V_0^2 a'(1 + a')\lambda_r^2.$$

$$16a^3 - 24a^2 + (9 - 3\lambda_r^2)a + (\lambda_r^2 - 1) = 0,$$

$$a = 1 - \frac{u}{V_0} \quad a' = \frac{1 - 3a}{4a - 1}$$

Axial Induction Factor correction



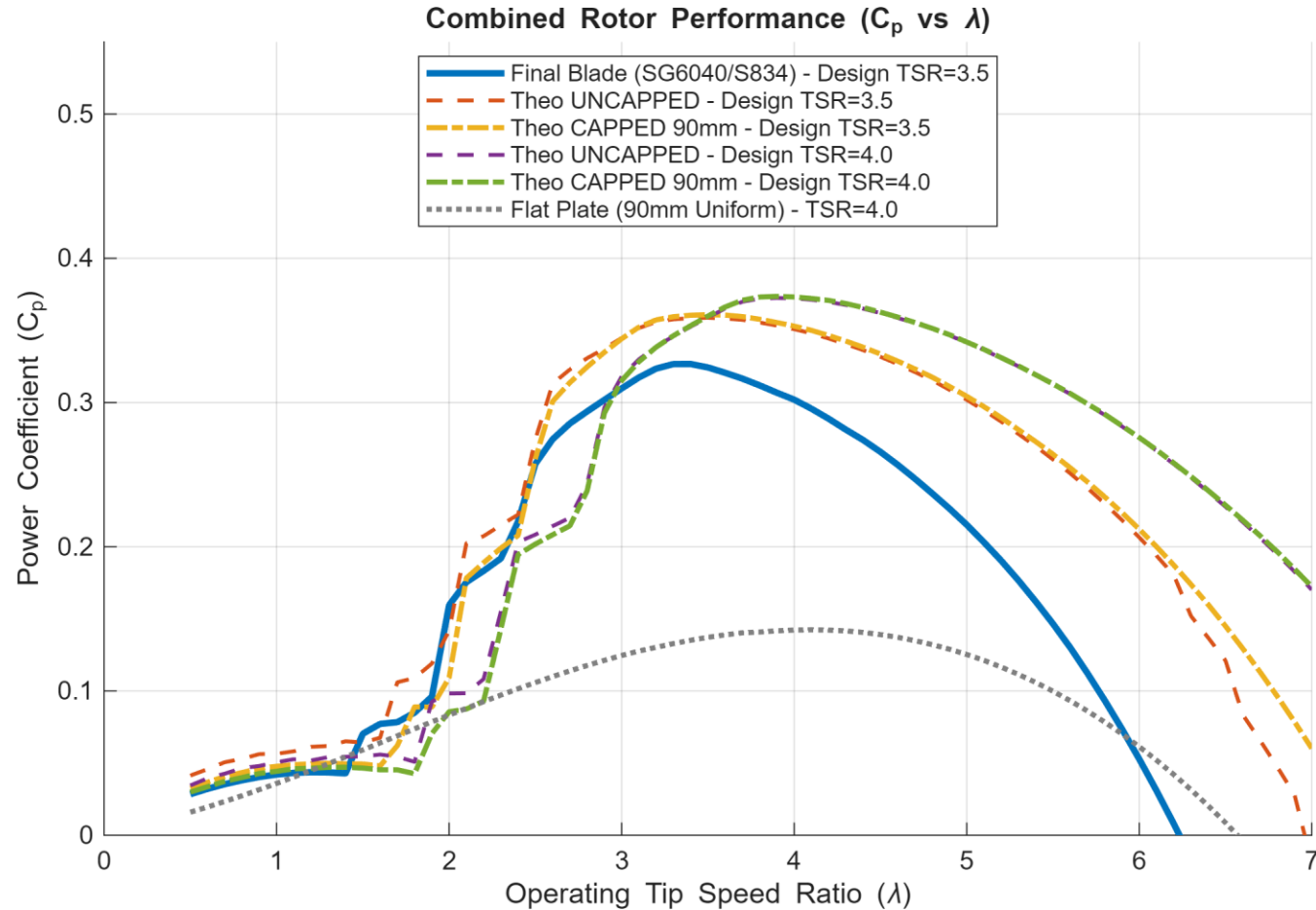
Tip Loss Correction

$$F_{tip} = \frac{2}{\pi} \arccos \left(e^{-\frac{B(R-r)}{2r \times \sin(\varphi)}} \right)$$

Root Loss Correction

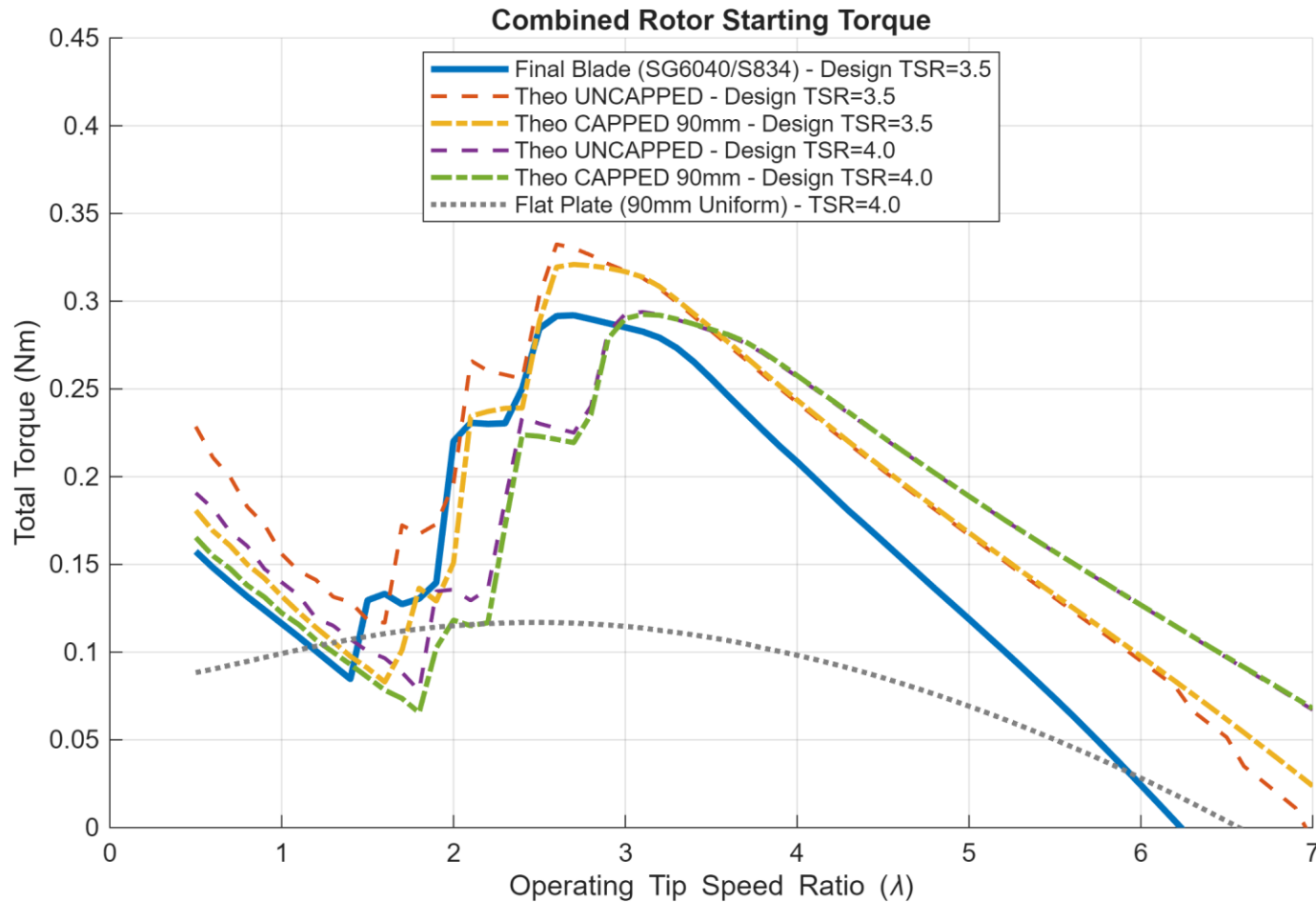
$$F_{root} = \frac{2}{\pi} \arccos \left(e^{-\frac{B(r-r_{hub})}{2r \times \sin(\varphi)}} \right)$$

Aerodynamic Performance Prediction



	Radius (m)	Peak C_p	Max Power (W)
Final Blade (TSR 3.5)	0.2430	0.3267	37.1210
Theo UNCAPPED (TSR 3.5)	0.2430	0.3588	40.7641
Theo CAPPED (TSR 3.5)	0.2430	0.3609	41.0058
Theo UNCAPPED (TSR 4.0)	0.2430	0.3725	42.3260
Theo CAPPED (TSR 4.0)	0.2430	0.3736	42.4510
Flat Plate (TSR 4.0)	0.2430	0.1424	16.1834

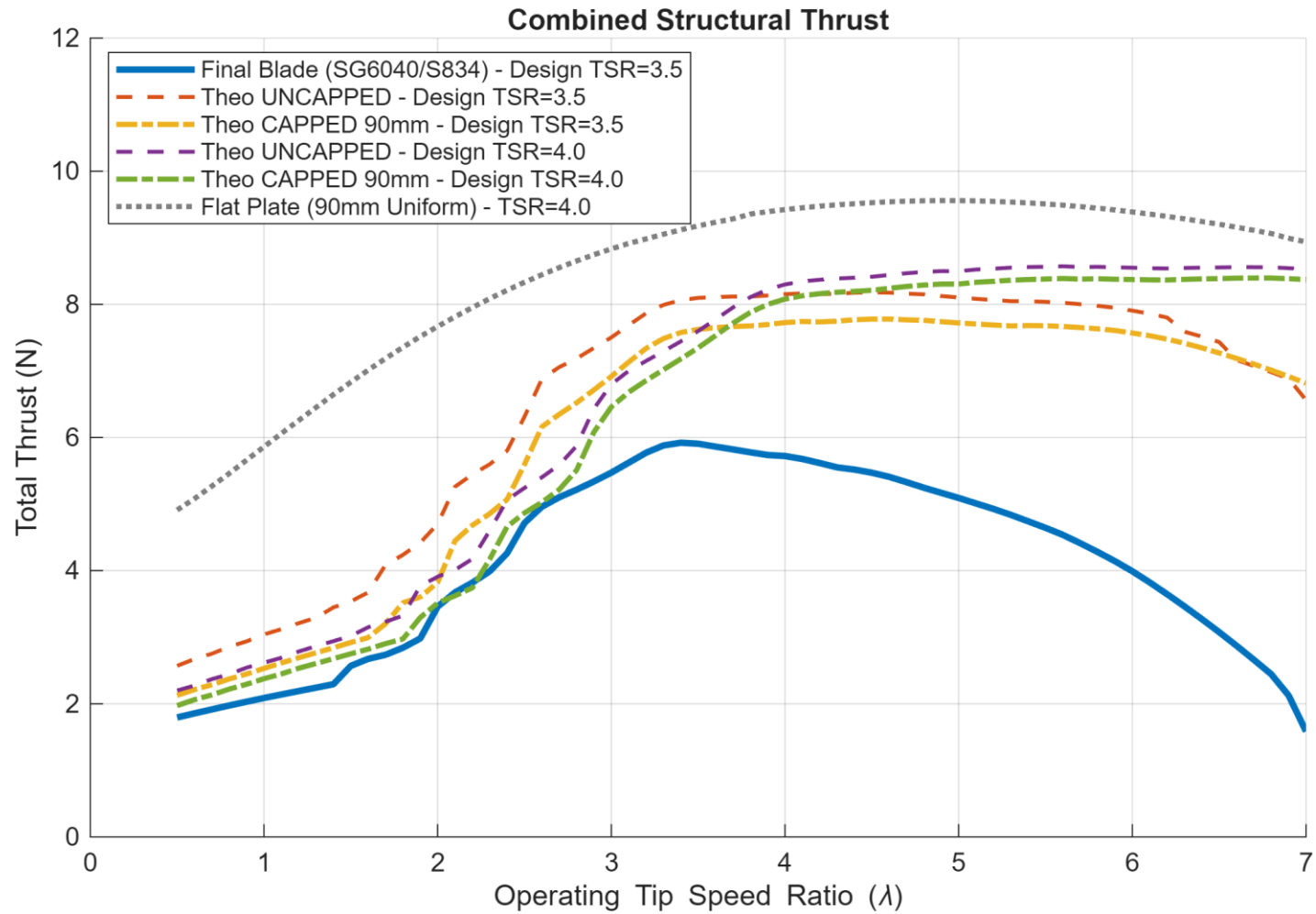
Aerodynamic Performance Prediction



	Minimum Spin-Up Torque (Nm)	Peak Torque (Nm)
Final Blade (TSR 3.5)	0.0847	0.2920
Theo UNCAPPED (TSR 3.5)	0.1167	0.3324
Theo CAPPED (TSR 3.5)	0.0829	0.3210
Theo UNCAPPED (TSR 4.0)	0.0780	0.2939
Theo CAPPED (TSR 4.0)	0.0655	0.2924
Flat Plate (TSR 4.0)	0.0884	0.1169

Aerodynamic Performance Prediction

Thrust by TSR Output



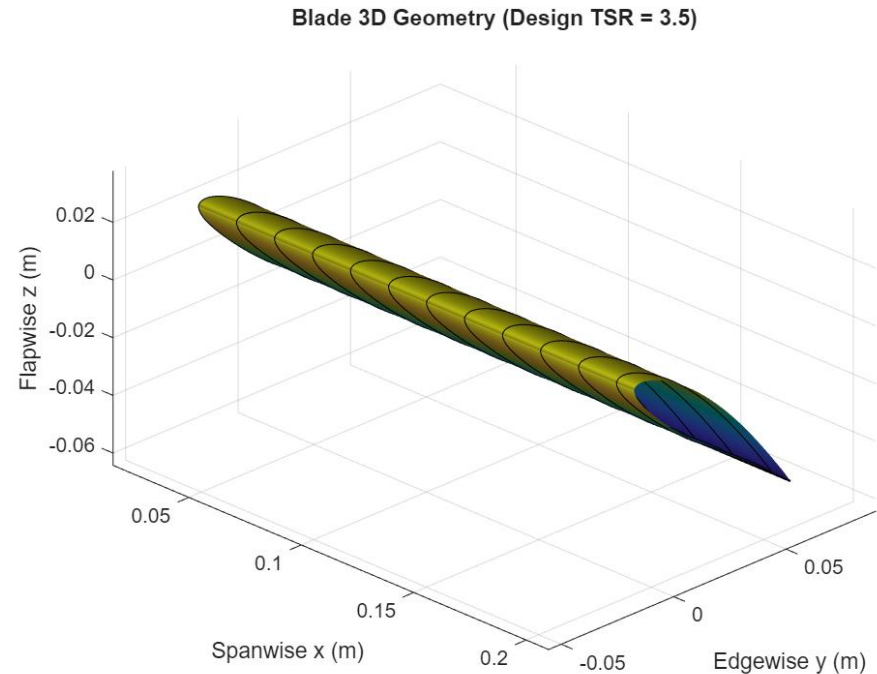
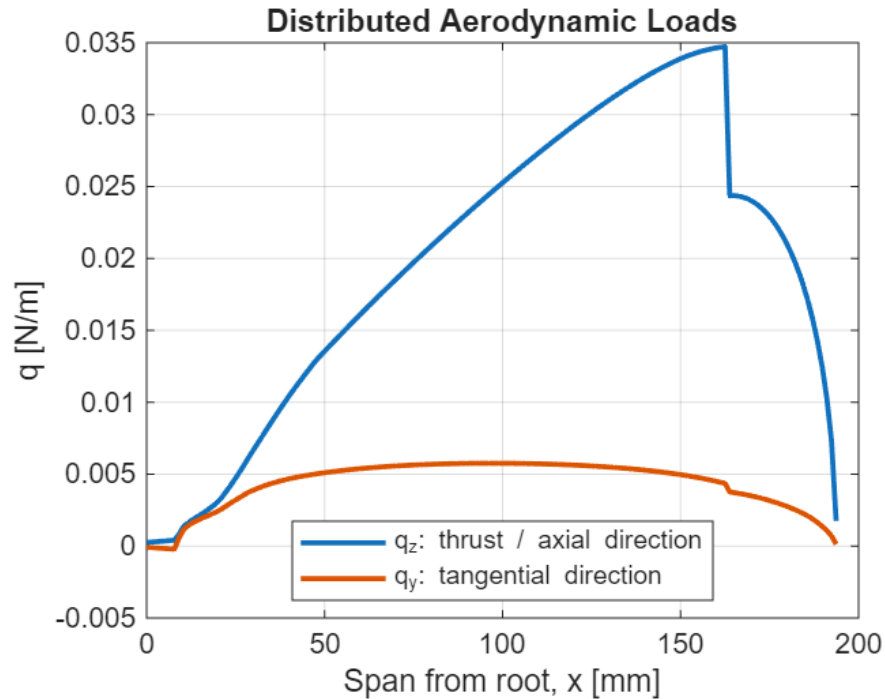
Aerodynamic Performance Prediction

Model Limitations

1. Design TSR yields sensitivity of the model between Betz and Glauert Optimum
2. Stall Correction – use of empirical constant is potential source of quantitative uncertainty
3. Hard switch of momentum model to empirical curve at $a=0.4$
4. Reynolds Snapping outside of 20k-100k range

Structural Performance Prediction

Blade Geometry and Loadings



Aerodynamics forces from BEM

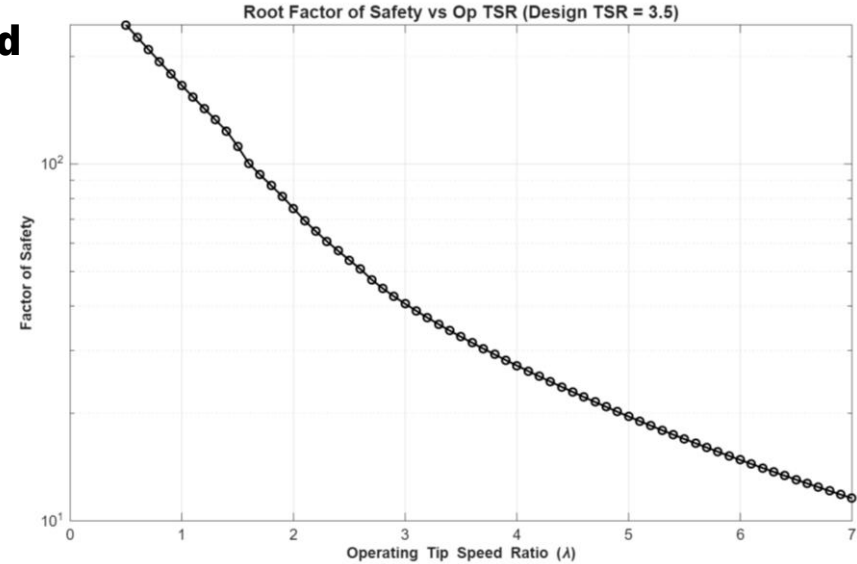
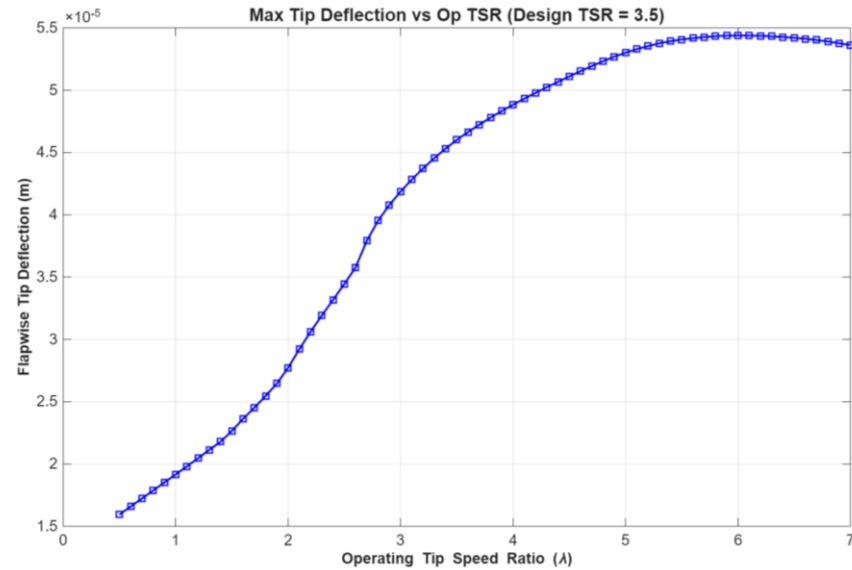


Converted spanwise distributed force for structure analysis

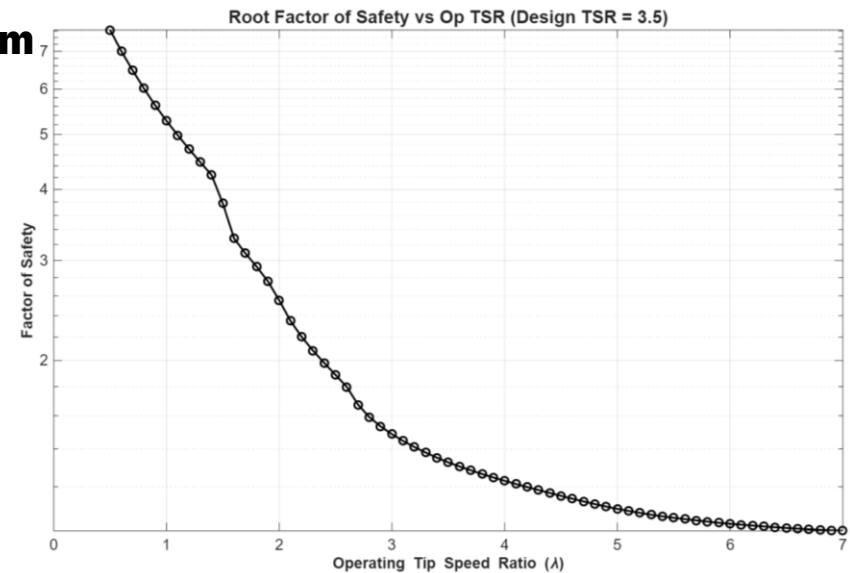
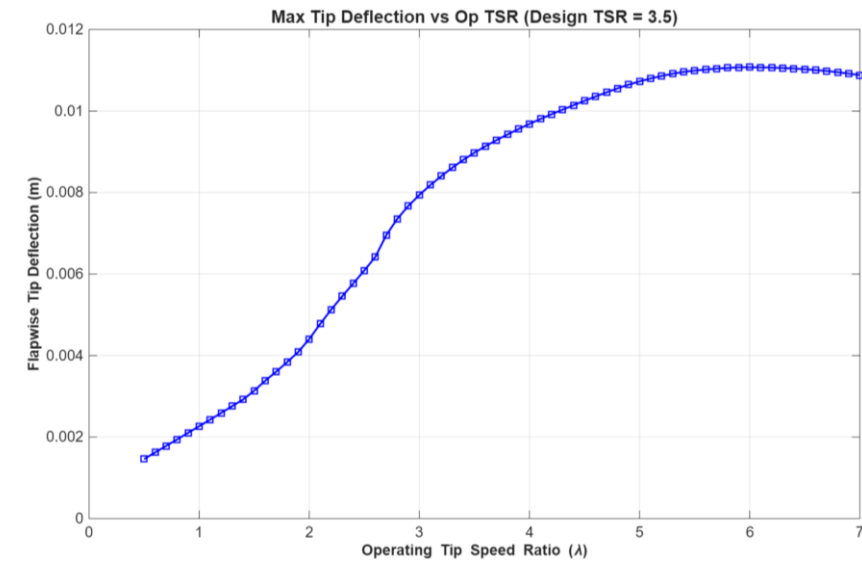
Structural Performance Prediction

Material Strategy

Plywood

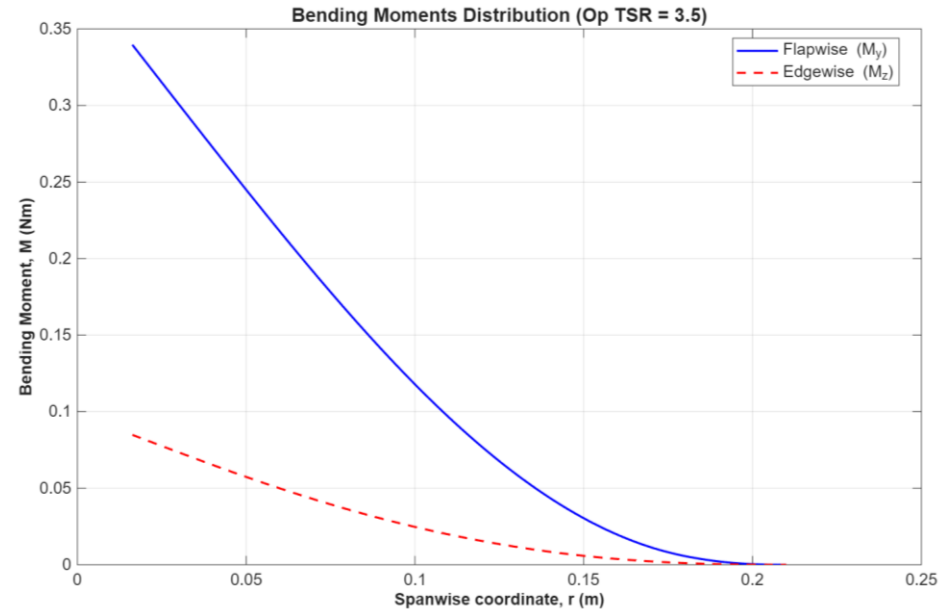
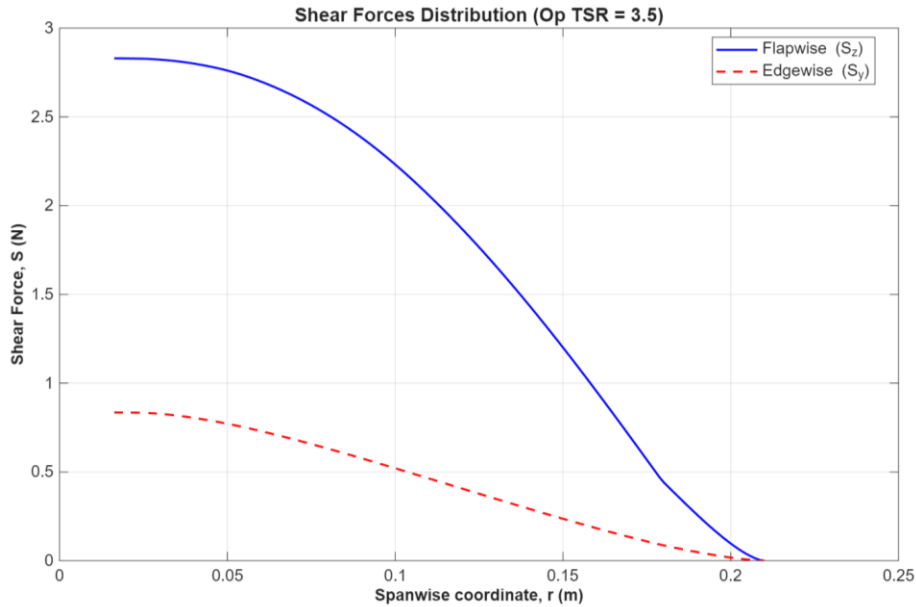


Styrofoam



Structural Performance Prediction

Forces and Bending Moments



$$T(x) = \int_x^R q(s) ds$$

Accumulated aerodynamic forces from tip

$$M(x) = \int_x^R T(s) ds$$

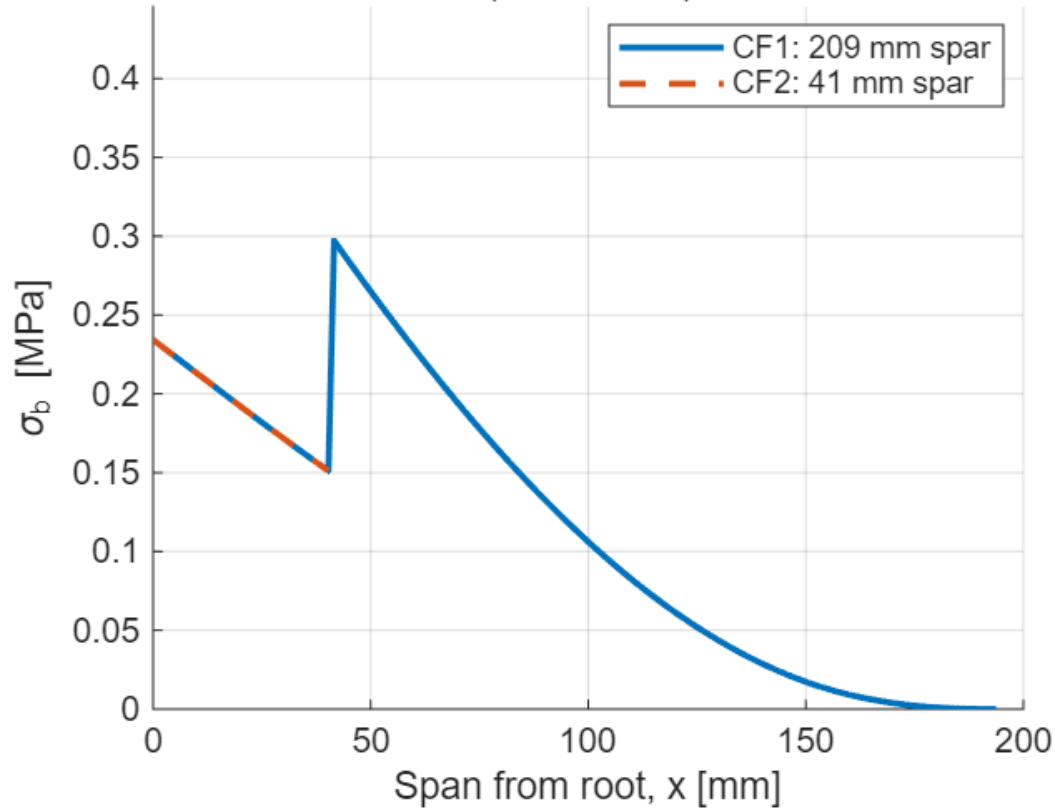
➡ **Maximum Bending Moment at Root**

Structural Performance Prediction

Simple Beam Analysis on Carbon Fibre Rod

Carbon Fibre Beam Bending Stress

Note: Limit (286.0 MPa) is off-chart



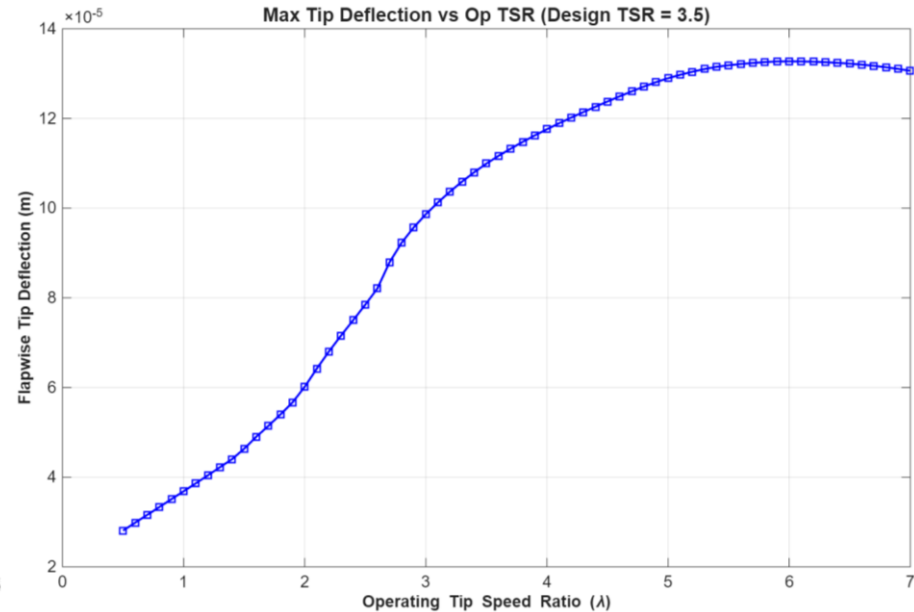
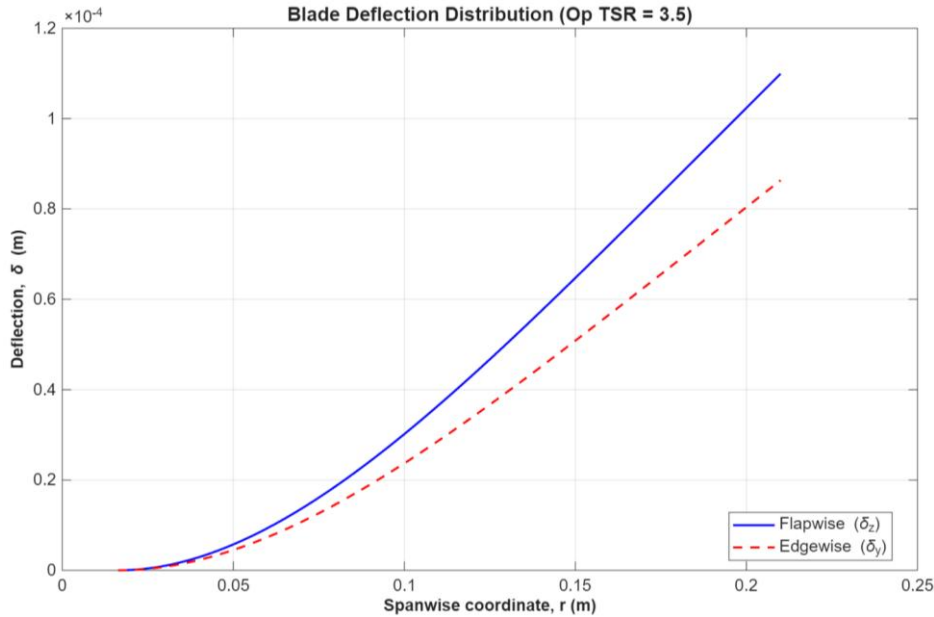
$$\sigma_x = \left(\frac{M_1}{I_1} \right) z_1 - \left(\frac{M_2}{I_2} \right) z_1$$

Carbon Fibre Only.

Neglect other materials, assuming CF resists all load.

Structural Performance Prediction

Deflection and Tower Clearance



Deflection for multi materials blade

Averaged E and ρ

$$E_{ave} = \frac{\sum_i (E_i \times I_i)}{I_{total}} \quad \rho_{ave} = \frac{\sum_i (\rho_i \times A_i)}{A_{total}}$$

$$EI \frac{d^2 \delta}{dr^2} = M(r)$$

$$\Rightarrow \delta(r) = \int \int \frac{M(r)}{EI} dr^2$$

Structural Performance Prediction

Conclusion

Material Selection

Foam/Plywood/PLA → Aerodynamical shape

Carbon Fibre Strips → Support majority of load

Structural Stability

Factor of Safety ~ 15.5

Tower Clearance/Deflection ~0.1 mm

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=====
                        STRUCTURAL ASSESSMENT REPORT - DESIGN TSR 3.5
=====

[1] MATERIAL & BLADE SPECIFICATIONS
-----
Material Assumption : Composite equivalent
Young's Modulus (E) : 1.84 GPa
Density (rho)       : 350.9 kg/m^3
Yield Strength      : 14.48 MPa
Fatigue Limit       : 6.07 MPa
Blade Length (R)    : 0.235 m

[2] SPANWISE DISTRIBUTION (At Rated Condition: Op TSR = 3.5)
-----
* Max Flapwise Shear (V_z) : 2.83 N (at Root)
* Max Flapwise Moment (M_y) : 0.34 Nm (at Root)
* Max Tip Deflection (z)   : 0.11 mm

[3] OPERATING TSR SWEEP (Worst-case Criteria Evaluation)
-----
> CRITERIA 1: Ultimate Strength (Target FoS >= 2.0)
* Yield Strength Limit : 14.48 MPa
* Maximum Peak Stress  : 0.93 MPa
* Critical Location     : r = 0.016 m
* Critical Azimuth     : 90 deg
* Critical Op. TSR      : 7.0
RESULT: [PASS] Minimum FoS is 15.53.

> CRITERIA 2: Tower Clearance (Limit <= 4.7 mm)
* Max Deflection Limit : 4.7 mm
* Maximum Deflection   : 0.1 mm
* Critical Op. TSR      : 6.0
RESULT: [PASS] Deflection is within limits.

> CRITERIA 3: Fatigue Limit
* Fatigue Limit        : 6.07 MPa
* Max Amplitude Stress : 0.02 MPa
* Critical Location     : r = 0.016 m
* Critical Op. TSR      : 5.0
RESULT: [PASS] Fatigue is safe.

=====
End of Report.
```

Prototype Manufacturing Methodology

Manufacturing Process Flowchart

Laser Cutting

3D print Root & Tip

Cut Carbon fibre

Glue Wood Sections

CF into Foam

Apply Adhesives

Assembly

Step 1 ➡ **Step 2** ➡ **Step 3** ➡ **Step 4** ➡ **Step 5** ➡ **Step 6**

Form Trailing Edge

Saw

Dremel & Sand

Cut Foam

Prepare for Adhesives

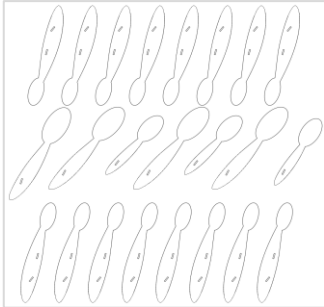
Surface Finishing

Final Sanding

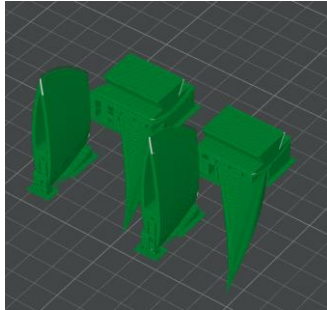
Prototype Manufacturing Methodology

Manufacturing Process photos

Laser-Cutting



3D-Printing



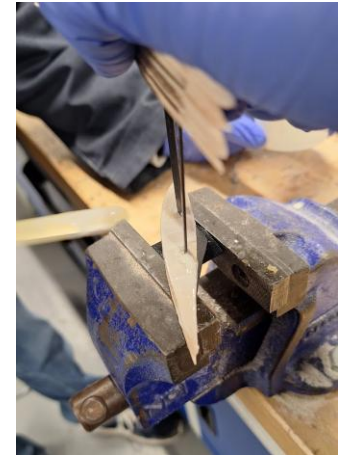
Sanding Wood



Cutting Foam



Assembling & Applying Finish



Prototype Manufacturing Methodology

Dimension & Aerofoil Shape Check



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Thank you

Presentation Title

DD/MM/YYYY

References

1.

Wing Bending Calculations. <https://web.mit.edu/16.unified/www/archives%202007-2008/spring%20temps/psets/systems/beambend.pdf> [Accessed 12th June 2026].

2.

Hau E. *Wind Turbines: Fundamentals, Technologies, Application, Economics*.. 3rd ed. Berlin, Heidelberg: Springer Berlin / Heidelberg; 2013.

3.

Ingram G. Wind Turbine Blade Analysis using the Blade Element Momentum Method. Version 1. 2011;

4.

Burton T, Jenkins N, Sharpe D, Bossanyi E, [eds]. *Wind energy handbook*.. 2nd ed. Chichester, West Sussex: John Wiley & Sons, Incorporated; 2011.

5.

Manwell JF, McGowan JG, Rogers AL. *Wind Energy Explained: Theory, Design and Application*.. 2. Aufl. s.l.: Wiley; 2010.

6.

Jamieson P. *Innovation in wind turbine design*.. 2nd ed. Hoboken: Wiley; 2018.

7.

Tyagi D, Schmitz S. Glauert's optimum rotor disk revisited – a calculus of variations solution and exact integrals for thrust and bending moment coefficients. *Wind Energy Science*. 2025;10(2): 451–460. <https://doi.org/10.5194/wes-10-451-2025>.

References

8.

Matsuoka K, Yoshida S, Muraoka Y, Yoshimizu H. Development of Stall Delay Built-In Actuator Line Model (SD-ALM) for Wind Turbine Rotor CFD. *Energies*. 2026;19(5): 1260.

<https://doi.org/10.3390/en19051260>.

9.

Liu S, Janajreh I. Development and application of an improved blade element momentum method model on horizontal axis wind turbines. *International Journal of Energy and Environmental Engineering*. 2012;3(1): 30. <https://doi.org/10.1186/2251-6832-3-30>.

10.

Yu G, Shen X, Zhu X, Du Z. An insight into the separate flow and stall delay for HAWT. *Renewable Energy*. 2010;36(1): 69–76. <https://doi.org/10.1016/j.renene.2010.05.021>.

11.

Hansen MOL. *Aerodynamics of Wind Turbines*.. 3rd ed. London: Taylor & Francis Group; 2015.

12.

Moriarty PJ, Hansen AC. *AeroDyn Theory Manual*. Report number: NREL/TP-500-36881, 2005 Jan. <https://doi.org/10.2172/15014831>.